

Panagiotis Zachos

SENIOR AUDIO & DISTRIBUTED SENSING ENGINEER · AI RESEARCHER

+306948226940 | panzaxos@gmail.com | Panagiotis-Zachos | panagiotis-zachos

Education

University of Patras

PhD in Signal Processing

Patras, GR

Sep 2020 – Nov 2024

- Thesis Subject: Sound Perception Optimization for Headphone Listening
- Summary: Development and optimization of algorithms that utilize auditory mechanisms and advanced digital processing methods aiming to achieve improved quality and naturalness in the perception of sound.

University of Patras

Integrated Master in Electrical and Computer Engineering

Patras, GR

Sep 2014 – Aug 2020

- Thesis: Sound Source Localization Using Deep Neural Networks Utilizing Binaural Cues
- Grade Point Average: **7.45 / 10**

4th General High School

High School

Larisa, GR

Sep 2011 – Jun 2014

- Graduated with Distinction
- Grade Point Average: **18.8 / 20**

Work Experience

Satways Ltd.

Senior Audio & Distributed Acoustic Sensing Engineer

Athens, Greece

Sep 2025 – Present

- Lead development of detection, DoAE, and classification algorithms leveraging Φ -OTDR technologies and distributed sensor arrays.
- Orchestrate experiments with microphone, hydrophone, and geophone arrays alongside Φ -OTDR DAS fibers.
- Build interactive geospatial dashboards (AIS, GNSS) to plan large-scale field trials.

Satways Ltd.

Audio & Distributed Acoustic Sensing Engineer

Athens, Greece

Dec 2023 – Aug 2025

- Engineered sensor arrays in multiple configurations and geometries for long-range surveillance.
- Delivered robust land/sea event detection and classification on Φ -OTDR DAS and acoustic arrays, combining classical signal/image processing with modern AI.

ITHACA Horizon Europe Project

Trustworthy AI & ML Researcher

Remote

Jun 2023 – Dec 2025

- Design systems that ensure trustworthy and ethical AI operation.
- Build tooling for AI cybersecurity, fairness, and privacy, with an accompanying visualization component.
- Create gamification and argument-visualization modules for a civic engagement platform.

Sonova Holding AG

Internship in Research & Development for Hearing Performance

Zurich, Switzerland

Nov 2020 – Apr 2021

- Prototyped and implemented time-domain signal-processing features in a new architecture.
- Conducted acoustic measurements.
- Built internal tooling for rapid integration and verification.
- Tuned and parameterized existing signal-processing functionalities.
- Ran hearing-performance experiments and validation.

PROMETHEUS Project, University of Patras

Machine Learning Researcher for Audio-Visual Signal Processing

Patras, Greece

Sep 2020 – Dec 2023

- Implemented a subsystem that generates static and dynamic 3D AR objects with interactive interfaces for AR-enhanced theatrical plays.
- Built a YOLO-based human pose-estimation module to track actors' movements in real time.
- Designed an intuitive GUI for directing and calibrating performances.

University Projects

YouTube Album Downloader

Personal

Github link

Nov 2019

- Command line application in Python to easily download full albums from YouTube and automatically split them in their respective songs by parsing the video description for song timestamps
- **Technical Skills**: Python, Shell, APIs.

Adaptive Filtering for Headphone Active Noise Control (ANC)

[Github link](#)

University of Patras

Jun 2019

- Design of 2 ANC applications utilizing adaptive filters, one with knowledge of environmental noise, via a microphone placed outside the headphone enclosure and one without information about the noise.
- **Technical Skills:** Signal Processing, MATLAB

Blood Pressure Monitoring System

[Github link](#)

University of Patras

Jun 2018

- An application written in Java consisting of two parts, a Blood Pressure Monitoring device (Raspberry Pi) with the role of the client, and a multi-threaded Server simulating a doctor's computer. The client automatically submits a patients' measurements to the server from where the doctor can remotely monitor their health.
- **Technical Skills:** Java, Raspberry Pi, Networking, Parallel/Threaded Programming.

Intelligent Control of a ZEB with FCMs

Patras, Greece

University of Patras

Jun 2018

- Design and modeling of a Zero Energy Building's power usage and production systems.
- Fuzzy Control Maps were employed to calculate the optimal energy balance to minimize carbon footprint.
- **Technical Skills:** MATLAB, Fuzzy Control Maps.

Control System and PIDF Design

Patras, Greece

University of Patras

Feb 2017

- Implementation of a transfer function (Plant) and a PIDF controller using LM741 Op-Amps.
- Performed simulations of the electrical circuits with PSpice and MATLAB Simulink to select the necessary components, laboratory measurements on the constructed circuit, and PCB designs via EAGLE Pcb for the Plant and PIDF Controller.
- **Technical Skills:** PIDF design, PCB design, Circuit simulations.

2048 Game Solver

[Github link](#)

University of Patras

May 2015

- Shell program written in C that automatically plays the popular mobile game "2048", deciding the best possible move through statistical analysis.
- **Technical Skills:** C, Statistical Analysis, Shell Game Development.

Skills

Audio Engineering

Digital Signal Processing, Active Noise Control, Headphone/Room Response Measurements, Room Equalization Wizard, Audacity, Reaper

Technical

Distributed Acoustic Sensing (DAS), Φ -OTDR, Phased Array processing, Software Defined Radio (SDR)

Programming

Python (NumPy, SciPy, Pandas, PyTorch, Tensorflow, matplotlib, parallel processing libraries), MATLAB, Simulink, git, C, Java, TI C6713 Assembly, C++

Platforms

Windows, Linux (Fedora, Debian, Ubuntu)

Honors & Awards

2018	2nd Place - EESTEC Challenge , EESTEC Challenge is an annual competition for engineering students with varying subjects. This competition involved processing Big Data and constructing Machine Learning models for planetary data.	Greece
2013	Euroscola Program , Accepted into the Euroscola program by winning the local round which involved writing an essay about the democratic deficit of the EU. During the program I was responsible for presenting a new ecological plan, devised by our team consisting of members from 5 different EU countries, in the European Parliament.	France
2006	1st Place - 4th Panhellenic Orchestra Competition , I was a member of the classical guitar ensemble from the 'Gkouletsou' conservatory, winning first place in the competition that took place in Piraeus University in Athens.	Greece

Publications

JOURNAL ARTICLES

Effectiveness of L2 Regularization in Privacy-Preserving Machine Learning

N. Chandrinos, I. Loi, P. Zachos, I. Symeonidis, A. Spiliotis, M. Panou, K. Moustakas

Under Review (2025), 2025

A multidisciplinary analysis of transparent AI-driven toxicity detection tools for civic engagement platforms

M. Zangl, I. Loi, P. Zachos, M. Bedek, E. Dimogerontakis, C.E. Nikolaou, D. Albert, K. Moustakas

AI & Society (2025) pp. 1–18, 2025

Targeted Beamforming Active Noise Control Based on Disturbance Metrics
Panagiotis Zachos, Georgios Moiragias, John Mourjopoulos
Acta Acustica (2024). 2024

Feedforward Headphone Active Noise Control Utilizing Auditory Masking
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
(2023). 2023

Low Filter Order Digital Equalization for Mobile Device Earphones
Gavriil Kamaris, Panagiotis Zachos, John Mourjopoulos
journal of the audio engineering society 69.5 (May 2021) pp. 297–308. 2021

CONFERENCE PROCEEDINGS

Beamforming Headphone ANC for Targeted Noise Attenuation
Panagiotis Zachos, John Mourjopoulos
Audio Engineering Society Convention 154, 2023, Helsinki, Finland

Targeted Beamforming ANC Based on Disturbance Metrics
Panagiotis Zachos, John Mourjopoulos
Forum Acusticum, 2023, Turin, Italy

Evaluation of Multichannel Reproduction of 3D audio in a Cave Automatic Virtual Environment
P Chatziantoniou, G Kamaris, K Kaleris, G Moiragias, P Zachos, S Agorgianitis, J Mourjopoulos
ACOUSTICS 2022, 2022, Thessaloniki, Greece

Binaural Data Reduction for Robust Sound Source Localization utilizing Convolutional Neural Networks for Reverberant Environments
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
ACOUSTICS 2022, 2022, Thessaloniki, Greece

Large Scale Headphone Response Analysis and Evaluation
Panagiotis Zachos, Dimitris Mermigas, John Mourjopoulos
ACOUSTICS 2022, 2022, Thessaloniki, Greece

An Efficient Data Reduction Method for Binaural Parameter Utilization
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
Forum Acusticum, 2020, Lyon, France

Languages

English	C2 - Advanced proficiency
Spanish	B2 - Professional proficiency
German	A2 - Basic proficiency
Greek	Native proficiency